

SPM Module 3: SKewed Data Generation

May 27, 2026

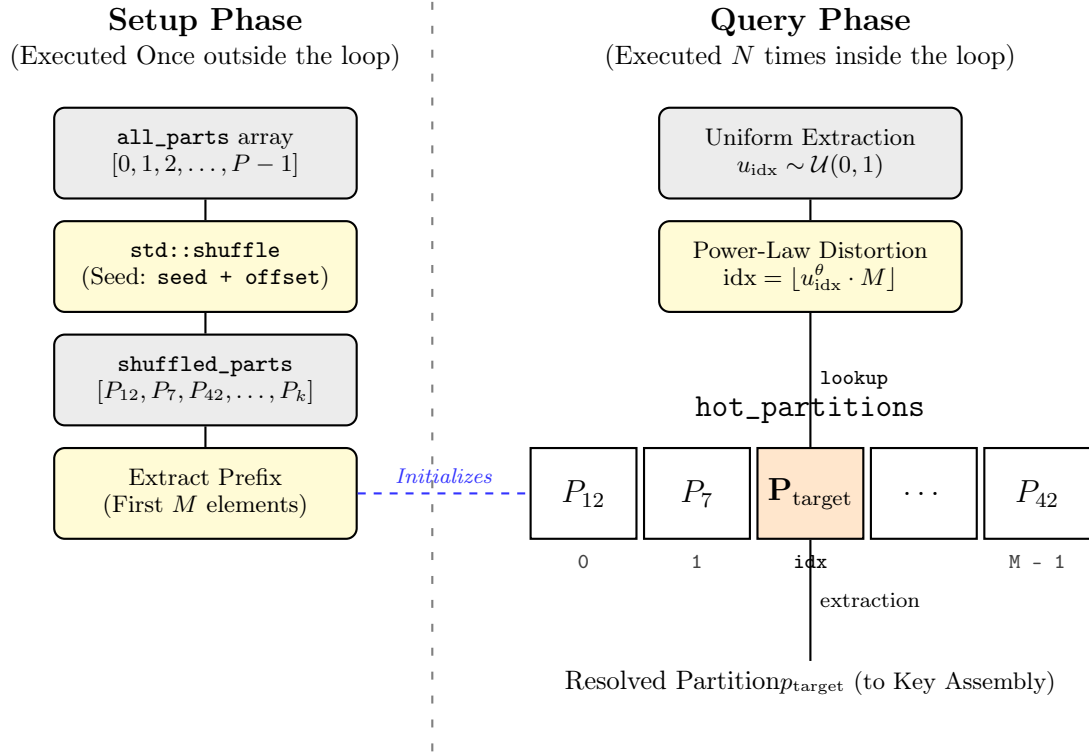


Figure 1: Lifecycle of the sparse partition subset. The `hot_partitions` array is populated once by shuffling the full partition space and extracting M elements. During generation, a power-law distortion with shape parameter θ is applied to a uniform random variable, skewing the index lookup towards the hottest partitions to resolve the physical ID (P_{target}).

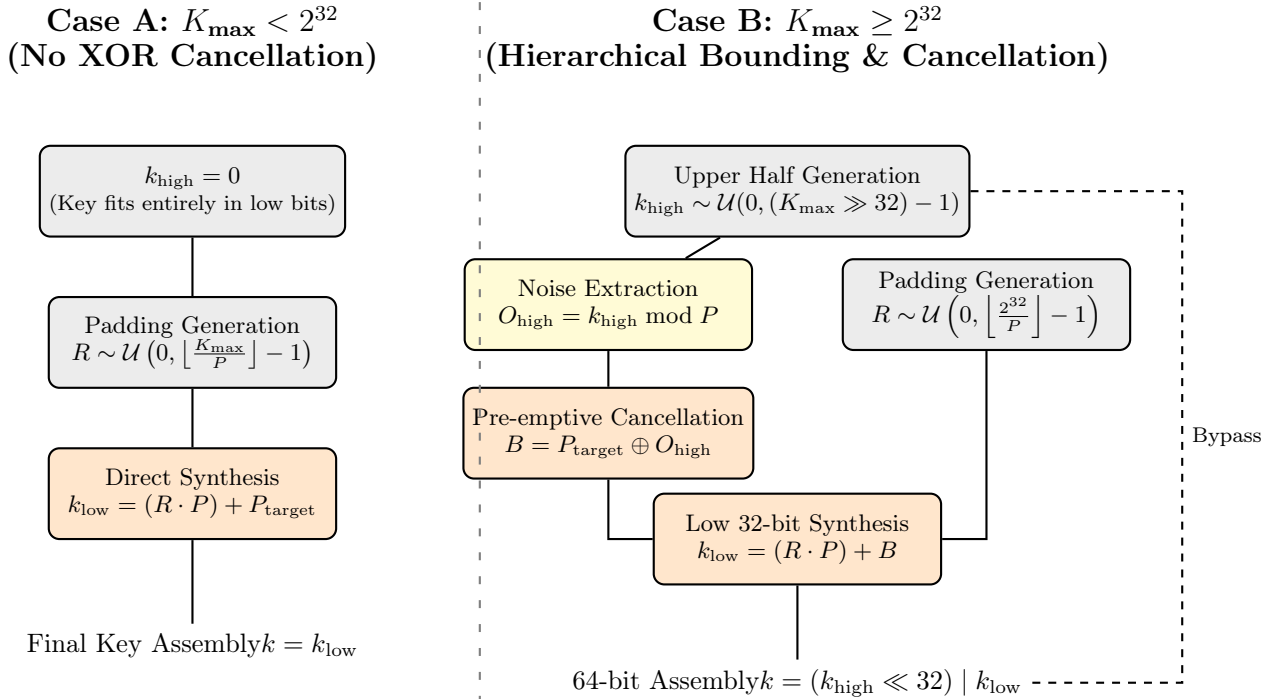


Figure 2: Key synthesis pipeline conditionally branching based on the global maximum key limit (K_{max}).

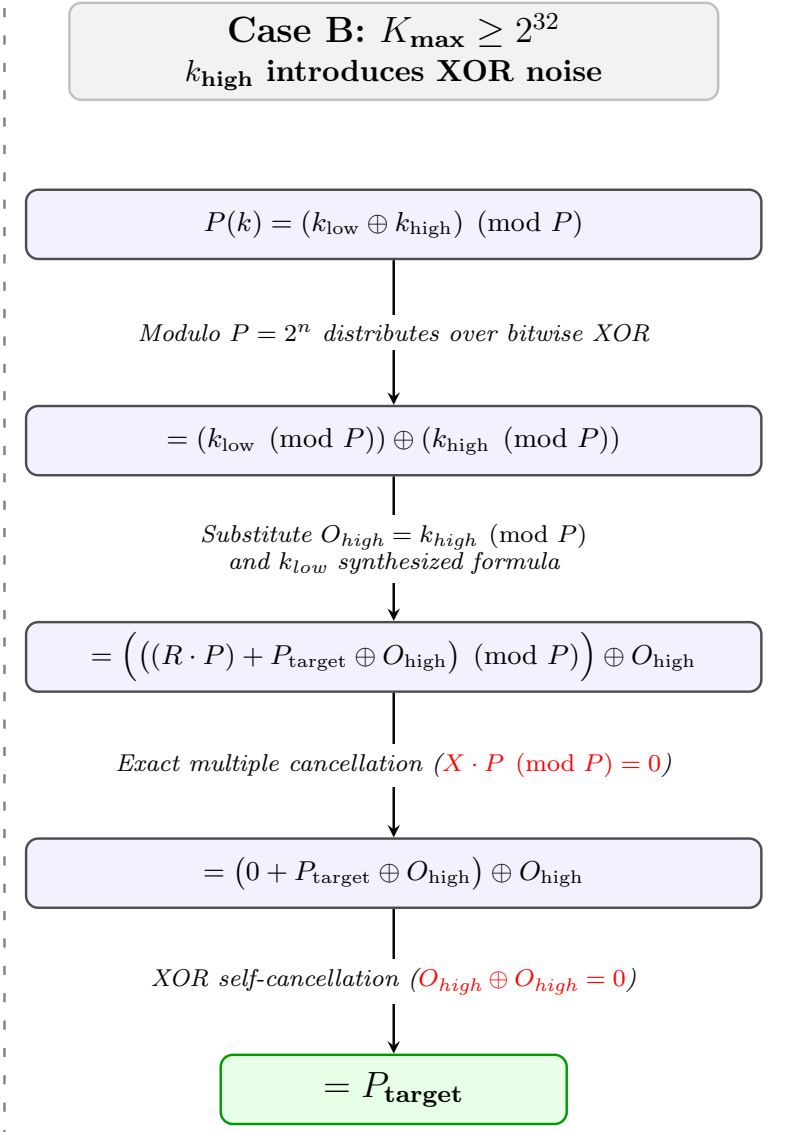
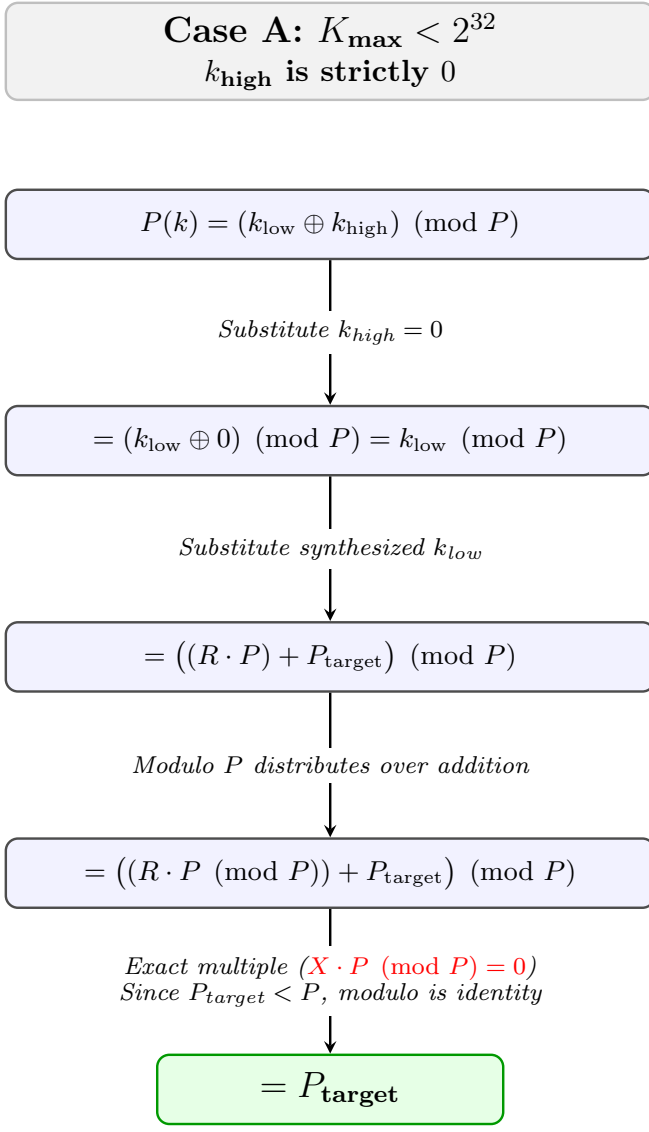


Figure 3: Mathematical derivation proving deterministic routing to P_{target} , adapting based on the bounds of $K_{\max}$.
 pgfplots